

Operations Research I

Part 1: Introduction to Optimization

IE 311

Catalog description. Linear and integer programming formulations; convex analysis; algorithmic design and the simplex method; duality and sensitivity; computer implementations. We will mainly focus on the following topics:

- LP/IP modeling
- LP solution methods – the **simplex algorithm** (requires convexity and linear algebra)
- LP theory, interpretations and advanced topics – **duality**
- **Implementation**

What is Operations Research?

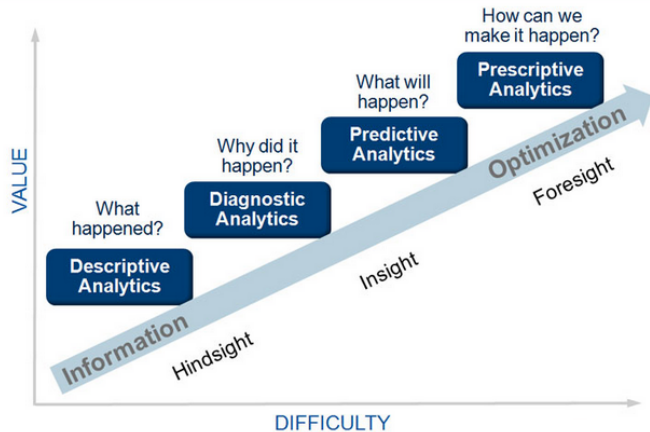
Operations Research is a scientific approach to decision making that seeks to design and operate a system *optimally*, usually under conditions requiring the allocation of scarce resources.

A **system** is an organization of interdependent components that work together to accomplish a certain goal.

The term *operations research* was coined during World War II, when scientists and engineers were asked to analyze several military problems.

Nowadays, the term **prescriptive analytics** is commonly used to mean "business practices utilizing optimization / operations research techniques."

Analytic Value Escalator



As we move up the value escalator, both the **value** and the **difficulty** of the analysis increase:

- **Descriptive analytics** – *What happened?* (hindsight)
- **Diagnostic analytics** – *Why did it happen?* (insight)
- **Predictive analytics** – *What will happen?* (foresight)
- **Prescriptive analytics** – *How can we make it happen?* (optimization)

Operations research lives at the top of the escalator: **prescriptive analytics / optimization**.

The scientific approach to decision making – that is, operations research – requires the use of **mathematical models**.

A **mathematical model** is a mathematical representation of the actual situation that may be used to make better decisions or to clarify the situation.

Optimization models *prescribe* the behavior for an organization that will enable it to meet its goals optimally. The components of such a model are:

- **Parameters** – what are given?
- **Decision variables** – what can be decided / changed?
- **Objective function** – what is the aim?
- **Constraints** – what should be obeyed / respected?

An optimization model seeks values of the decision variables that **optimize** (maximize or minimize) an objective function among the set of all values for the decision variables that **satisfy the given constraints**.

- Trying to write the model before clearly **defining the decision variables**.
- Mixing up the **parameters** (given in the problem definition) and the **decision variables** (which should be defined by the modeler!).
- Making assumptions about how variables would behave – this is often difficult and may lead to incorrect formulations.
- **Not relating decision variables properly** – all variables should be somehow "connected" to each other.
- Ignoring certain aspects of the problem, such as omitting constraints (including variable domains).

With respect to the relations between decision variables:

- *Linear* – objective and constraints are linear functions.
- *Nonlinear* – some component of the objective and/or constraints is nonlinear.

With respect to the types of decision variables:

- *Continuous* – all variables are real numbers.
- *Integer* – all variables are integers.
- *Mixed-integer* – some variables are continuous, the others are integers.

With respect to the characteristic of parameters:

- *Deterministic* – all parameters are known. (*We focus only on these.*)
- *Stochastic / uncertain* – some parameters are random / uncertain.

Types of Linear Mathematical Optimization Models

An optimization problem with linear relations between **continuous** variables is a **Linear Program (LP)**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0$$

An optimization problem with linear relations between **integer** variables is a **(Pure) Integer Linear Program (IP)**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \quad \text{and} \quad x_1, x_2 \text{ integer}$$

An optimization problem with linear relations between **continuous and integer** variables is a **Mixed-Integer Linear Program (MILP)**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \quad \text{and} \quad x_1 \text{ integer}$$

Types of Nonlinear Mathematical Optimization Models

An optimization problem with nonlinear relations between **continuous** variables is a **Nonlinear Program (NLP)**:

$$\max; 2|x_1 - 1| + 3x_2 \quad \text{s.t.} \quad x_1 + x_2^2 \leq 5, ; ; x_1, x_2 \geq 0$$

An optimization problem with nonlinear relations between **integer** variables is a **(Pure) Integer Nonlinear Program (INLP)**:

$$\max; 2|x_1 - 1| + 3x_2 \quad \text{s.t.} \quad x_1 + x_2^2 \leq 5, ; ; x_1, x_2 \geq 0 \quad \text{and} \quad x_1, x_2 \text{ integer}$$

An optimization problem with nonlinear relations between **continuous and integer** variables is a **Mixed-Integer Nonlinear Program (MINLP)**:

$$\max; 2|x_1 - 1| + 3x_2 \quad \text{s.t.} \quad x_1 + x_2^2 \leq 5, ; ; x_1, x_2 \geq 0 \quad \text{and} \quad x_1 \text{ integer}$$

We will focus on the modeling of the following classes of optimization problems (and we will also solve them with software!):

- **Linear Programming**
- **(Mixed-) Integer Linear Programming**
- Special cases of (Mixed-) Integer **Nonlinear** Programming that can be expressed as (Mixed-) Integer Linear Programming – for example, when the nonlinearity is due to a piecewise-linear function.

We will then focus on the **solution methods** (i.e. the simplex method) and the **theory** (e.g. the structure of polyhedra and duality) of Linear Programming.

- You will learn solution methods for Mixed-Integer Linear Programming, Network Models, Dynamic Programming (and many more!) in **IE 312**.
- You will learn some stochastic models in **IE 313**.
- Many IE courses (304, 401, 408, 409, 414, 415, 430, 454, 471) are built upon optimization knowledge.
- You may use optimization in your **internship project**.
- You are very likely to use optimization in your **senior design project**.

One can almost say that

Industrial Engineering = Optimization + Probability/Statistics + Modeling + Computing.

- Optimization / analytics knowledge can give you a **competitive advantage** over other engineers in the job market.
- You need to know optimization very well if you would like to pursue **graduate studies** in OR / IE / Management, especially at the PhD level.

- Attend all lectures (2 hrs), labs (1 hr) and recitations (2 hrs).
- Study for another **5 hours every week** on your own time.
- Do not only read the solutions – work on the questions yourself (or with your classmates), *then* check the solutions.
- **Write your own code.**
- Do not miss quizzes.
- Prepare for exams very well.
- Embrace the fact that this course requires a lot of work!